

— SONG YOUNG SHIN DAIRY FARM · REPUBLIC OF KOREA

Sustainable Dairy Farming

Smart Farming (IT · BT) & Circular Agriculture

스마트팜과 순환농업으로 만드는 지속가능한 낙농

Xin chào — 베트남 대표단을 환영합니다 · **Chào mừng đoàn Việt Nam**

Hyunjae Ha, DVM 하현제 수의사

Goryeo VET Clinic · Song Young Shin Farm · Farm Study Visit 2026

A warm welcome to our friends from Vietnam

It is an honor to host public officials of Vietnam at our farm. We share one mission — building a livestock industry that is **profitable for farmers, kind to animals, and gentle on the land**. Today we show how a Korean family dairy turned that mission into daily practice.

송영신목장을 찾아주신 베트남 공무원 여러분을 진심으로 환영합니다.

Built around your four areas of interest



Eco-friendly livestock

친환경축산



Circular agriculture

순환농업 · Nông nghiệp tuần hoàn



Sustainable dairy

지속가능한 낙농



Smart farm

스마트팜 · IT & BT

What we will walk through today

01 Who we are

The D2O ecosystem — farm, clinic, research, platform

03 Sustainable dairy

Song Young Shin Farm — welfare, low-carbon, grass-fed A2

02 Circular agriculture

순환농업 — the manure-to-soil nutrient loop & D2O

04 Smart farming (IT · BT)

smaXtec rumen sensors & genetics / embryo transfer

05 A model for Vietnam & Q&A

How this transfers to Vietnamese dairy — and open discussion

One integrated ecosystem for smart, sustainable dairy



L&H Research Center

- Sexed embryo transfer (MOET)
- IVF & embryo culture (BT)

Goryeo VET Clinic

- ET / AI & smart vet system
- Reproduction & feeding consulting

Song Young Shin Farm

- Animal-welfare & low-carbon certified
- A2 hay milk — Jersey &

D2O Co., Ltd

- Digital platform for dairy
- Smart-farming solutions (dairy & pig)

Six forces shaping sustainable agriculture

Profitability

수익성

A farm must earn to endure.

Climate change

탄소 · 메탄 · 농업

Low-carbon livestock is now the standard.

One Health

동물 · 자연 · 인간

Animal, environment & human health are one.

Smart farming

노동력 · 승계 · 삶의 질

Technology keeps the next generation on the farm.

Animal welfare

세계적 이슈

Comfortable cows are productive cows.

Humanity

철학 · 사명감

Farming as a calling, not only a business.

Waste is not waste — it is a resource in the wrong place

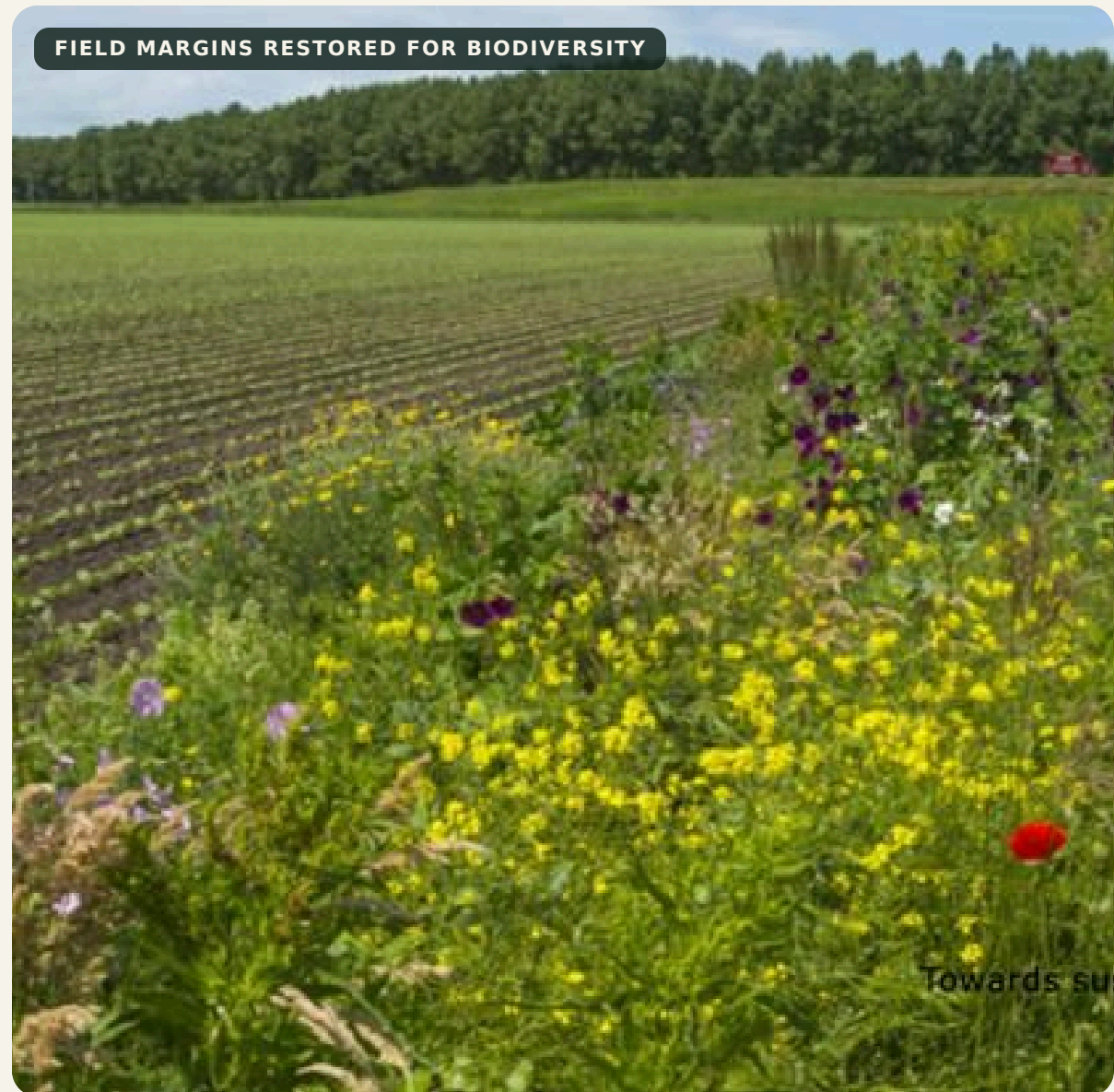
순환농업이란 무엇인가?

- **Manure → Compost → Feed crops → Cattle feed** — a closed nutrient loop
- An eco-economic system that turns a cost into an input
- A direct, practical response to climate change

↓ Feed costs

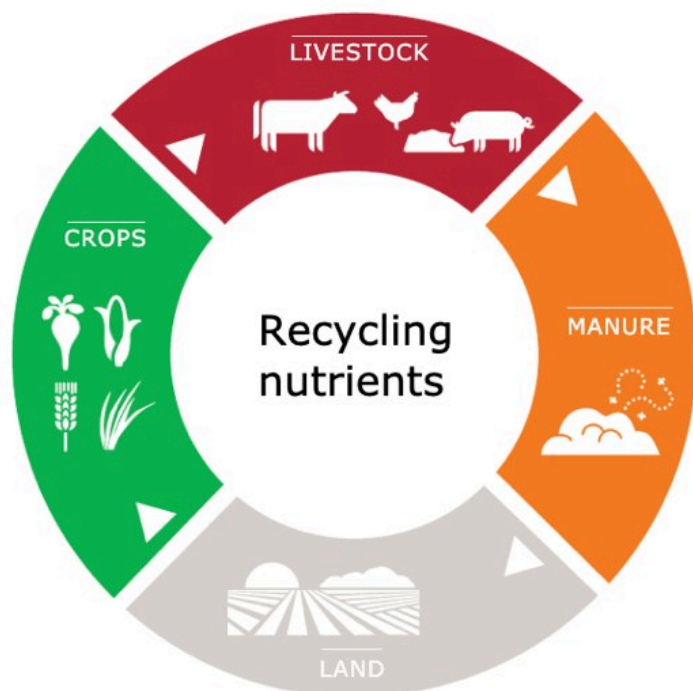
↓ Pollution & carbon

FIELD MARGINS RESTORED FOR BIODIVERSITY



Towards su

The livestock-crop nutrient cycle



1

Feed → Livestock → Manure

Nutrients circulate instead of leaking out of the system

2

Manure = organic fertilizer

Returned to the grassland that grows the herd's own feed

3

Low emission & sustainable

Fewer inputs bought, less pollution released

SONG YOUNG SHIN FARM — GYEONGGI, KOREA



D2O — returning dung to its origin, the soil

The heart of eco-friendly livestock is what you do with manure. D2O is our barn-and-bedding management approach that treats manure as the starting point of the next harvest — not a disposal problem.



Odor reduction

A cleaner barn for animals, workers and neighbours — critical near communities.



Healthier bedding

Drier, better-managed bedding supports udder health and animal comfort.



Low-carbon compost

Faster, controlled composting returns stable organic matter to the grassland.

Why it matters for Vietnam: in hot, humid climates manure & odor management is the first bottleneck to scaling dairy near towns. Solve it, and welfare, environment and community acceptance all improve together.

HBP — a peat-moss bedding that feeds the soil

The **HBP system** makes the barn floor the first step of the nutrient loop. A peat-moss-based bedding absorbs moisture and odour, keeps cows dry and healthy, then composts quickly into organic fertiliser for the grassland — closing the D2O circle.

High absorption

Peat moss locks in moisture & ammonia — a dry floor, far less odour.

Animal welfare

Soft, dry bedding → cleaner udders, healthier hooves & skin.

Pathogen control

A dry, low-pH bed discourages harmful bacteria in the barn.

Closes the loop

Spent bedding + manure compost fast → back to the soil as fertiliser.

For Vietnam: in a hot, humid climate moisture & odour are the barn's first problem — HBP is a direct, low-cost answer that improves welfare and the environment together.



Song Young Shin Farm

Korea's 1st animal-welfare & low-carbon certified dairy farm

1st Animal-welfare &
low-carbon certified

30^{ha} Own grassland for
home-grown feed

A2 Grass-fed hay milk
Jersey & Holstein



Robotic milking —
cows choose their time

THE FARM



GRASS-FED JERSEY HERD



30 HA GRASSLAND



A2 HAY MILK RANGE

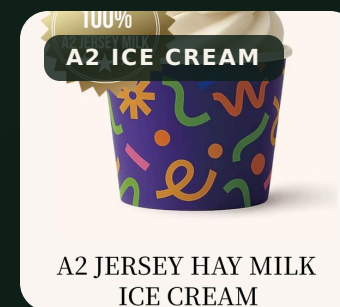


PRODUCT OF THE SYSTEM · A2 HAY MILK

No corn. No grain. Just grass.

A sustainable farm produces a distinctive product. Cows grazed and fed on the farm's own grassland give **A2 β -casein milk** — sold as a premium, traceable brand that funds the farm's continued investment in welfare and technology.

- ✓ 100% grass-fed — no corn, no grain
- ✓ A2 Jersey & Holstein hay milk
- ✓ Welfare-first: robotic milking, freedom of movement
- ✓ Farm-to-brand traceability that consumers trust



Hay milk · Yogurt · Ice cream — value added
on the farm

Biotechnology: better cows, faster

At L&H Research Center we use **embryo transfer (MOET)** and **IVF** to multiply the best dairy genetics — cows that give more milk, breed reliably, and cope with a changing climate.

- 1 **Oocyte recovery** from selected donor cows
- 2 **Maturation & in-vitro fertilization**
- 3 **Culture** to blastocyst, quality selected
- 4 **Transfer** to recipient — or freeze at -196°C

Focus: **Korean Holstein sexed dairy embryos** — improving milk herds sustainably.



RUMEN BOLUS SENSOR, INSIDE THE COW



— SMART FARMING · IT · SMAXTEC

One sensor. Three jobs. 24/7.

A small bolus rests in the cow's rumen and continuously measures **inner body temperature (± 0.01 °C)**, rumination and drinking — the earliest, most reliable signals of what a cow needs.



Health

Temperature, rumination
& water intake



Reproduction

Heat window & calving
prediction



Feeding

Ration-group rumination
& pH option

TruDrinking™ · TruRumi™ · TruAdvice™ — with automatic work lists & SOPs

Catching illness days before you could see it

Inner body temperature is the **earliest health signal** a cow gives. Mastitis and metritis show short fever bouts **2-5 days before** any clinical sign — enough time to act before the animal is sick.

2-5

days early warning

40-70%

potential savings via early care

AI — TruAdvice™

The system doesn't just show data — it flags likely conditions:

Mastitis

Ketosis

Milk fever

Every pregnancy, on time



Heat detection

- Reads activity, rumination, temperature & drinking together
- Heat index shows how strong the heat is
- Per-cow ideal insemination time — **12-hour window**



Calving prediction

- Calving alert **15-30 hours in advance**
- Be present for hard births — save cow & calf
- Two calving alerts flag raised **milk-fever risk**



The result

- Fewer missed heats, shorter calving interval
- Higher reproduction rate = more milk, more calves
- Less hormone use, less guesswork

Four transition-period diseases, caught early

The riskiest window is right after calving. Each disease has a temperature & rumination fingerprint the sensor reads before the cow looks ill.

Mastitis

3-5 days early

Several short temperature spikes lasting only hours — before milk yield or conductivity change.

Metritis / Endometritis

first 14 days

Prolonged fever periods early in lactation, days before clinical symptoms appear.

Milk fever

around calving

Temperature drop after calving; missed drinking & poor rumination flag cows at risk.

Ketosis

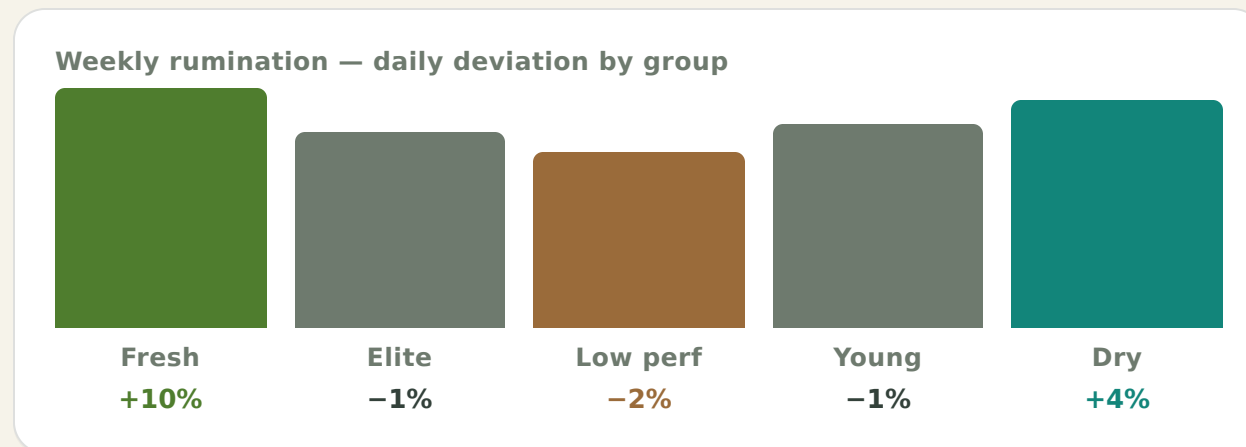
temp + rumination

Temperature & rumination both fall = clinical; often avoided if the earlier disease was caught.

Feeding, judged by the cow's own rumen

Rumination time is the truest measure of whether a ration is working. The system tracks **weekly rumination and daily deviation for every ration group** — fresh cows, high performers, dry cows, heifers.

- Spot a feeding problem the day it starts, not weeks later
- Balance rations by real digestion, not by eye
- Steady rumen = steady health, fertility & milk



Illustrative of the smaXtec feeding dashboard

WHY IT WORKS · BENEFITS

What smart farming delivers

- 1 **Early disease detection** — lower treatment cost
- 2 **Higher reproduction rate** — more productivity
- 3 **Labor saving** — 24/7 automatic monitoring
- 4 **Data-driven decisions** — scientific management

Fit for Vietnamese farms 🇻🇳

- ✓ Suited to 10-50 head family farms
- ✓ Works in hot & humid climates
- ✓ Simple smartphone monitoring
- ✓ Easy bolus installation, lifetime use per bolus
- ✓ Long-term return on investment
- ✓ Adapted from real Korean farm experience

From this farm to Vietnamese dairy — a shared path

01

Pilot

A demonstration herd with sensors, welfare housing and circular manure management — proof on Vietnamese soil.

02

Localize

Adapt to tropical climate, local feed and herd size; train veterinarians & farmers; build the data baseline.

03

Scale

Extend to cooperatives and provinces — a national platform for eco-friendly, data-driven dairy.

We offer **knowledge exchange, training and technology partnership** — bringing eco-friendly livestock, circular agriculture, sustainable dairy and smart-farm practice together for Vietnam.

OUR PHILOSOPHY · 농민의 세상은 무궁무진합니다

"The farmer
holds the key to
humanity's eternal life
vault."

— Yun Bong-gil, *The Farmer's Reader*, 1927

건강한 흙

→

행복한 소

→

좋은 우유

Soil to Soul

감사합니다

Cảm ơn quý vị · Thank you

It is an honor to share our work with you. May our farms grow together.

Hyunjae Ha, DVM 하현재 수의사

Goryeo VET Clinic · L&H Institute of Biotechnology

Song Young Shin Farm · D2O = Dung to Origin